

Appendix

Appendix A. Data construction

A.1 Panel set and dependent variables

In the main regressions, we use the number of popular uprisings occurring in each county-month as the core dependent variable. These data are primarily drawn from Aoki (1971), which provides a systematic compilation of populace resistance during the Tokugawa and early Meiji periods. This approach is consistent with existing studies on social conflict in Japan, such as Lei et al. (2025) and Steele et al. (2017). The dataset records various forms of collective resistance initiated by commoners in response to rising prices, policy grievances, and economic distress, including petitions, protests, and riots, rather than rebellions led by samurai elites.

An example of the data structure is shown in the figure below. Each column corresponds to a specific event, and each row records the following information: (1) the year of occurrence; (2-3) the geographic location at the province (*kuni*) and county (*gun*) levels; (4) domain affiliation (*han-ryō* or *bakufu shihai-chi*); (5) the village or town where the event took place, when available; (6) the stated cause of the resistance; (7) the form of organization; (8) the historical source; and (9) the corresponding modern place names.

Row 2 to 4 together identify the location of each event. Specifically, rows 2 and 3 denote geographic units, while row 4 indicates the governing authority under the feudal system, such as domain rule or direct shogunal control. This structure makes it feasible to aggregate conflicts at the county level. Counties represent relatively stable geographic units. This feature allows us to construct a county-level panel dataset by aggregating all events within each county. Information on domain affiliation is used in the analysis of institutional responses in Section 5 (see Appendix D for details).

Finally, row 7 distinguishes different forms of resistance. In this paper, we focus primarily on large-scale and more violent forms of conflict, such as violent attacks on merchants or local officials, large protests, and mass riots organized by the population. In Table 4 of the main text, we further examine more moderate forms of collective action, including petitions, dispersions, and intra-village disputes.

5 ・ 8	5 ・ 8	5 ・ 7	5 ・ 4	3 ・ 29	2 ・ 2	8 ・ 1	6 ・ 29	6 ・ 28	6 ・ 5	1 ・ 1	慶 応 (八云)	年 月 日	→	Year; Month; Date
摂津 川辺	摂津 武庫	信濃 下伊那	河内 石川	大和 宇陀	尾張 愛知	越後 中頸城	相模 高座	越後 南魚沼	伊豆 田方	加賀 石川	摂津 川辺	攝津 武庫	→	Province (<i>Kuni</i>); County (<i>Gun</i>)
近衛氏領	天領	飯田	天領	松山	名古屋	高田	天領	天領	天領	金沢	近衛氏領	天領	→	Domian (<i>Han</i>) or shogunal territories
伊丹町	兵庫	時又村・ 兵庫	富田林町	松山町	名古屋町	直江津町	藤沢町	塩沢町	三島町	金沢町	伊丹町	兵庫	→	Towns and villages
米蔵、貧民二千人、米酒商 など四一戸おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	米蔵、穀屋おそす	→	Popular resistances reason
わし	わし	わし	わし	わし	わし	わし	わし	わし	わし	わし	わし	わし	→	Popular resistances type
酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	酒井・前掲論文	→	Reference
伊丹市	神戸市	飯田市	富田町	大宇陀町	名古屋	直江津市	藤沢市	塩沢市	三島市	金沢市	伊丹市	神戸市	→	Current location

Figure A1: Example for original popular resistance data.

A.2 Climate shocks

Temperature shock data are drawn from the “Asia 1200-Year Gridded Summer Temperature Reconstructions” developed by Cook et al. (2013)¹. The dataset is constructed using a network of annual tree-ring chronologies across temperate East Asia. An ensemble version of point-by-point regression is applied to reconstruct summer temperatures at individual grid points. This procedure generates 585 grid-level summer temperature series covering the period 800-1989. The reconstructions are statistically calibrated and validated against instrumental records, ensuring reasonable reliability for regional temperature variation. Using the 1961-1990 average as the baseline, the dataset reports annual temperature anomalies for each grid cell in degrees Celsius. These grid-level reconstructions are then used to measure spatial variation in summer temperature shocks across regions. Figure 6 in the context provides an example of temperature anomaly for the year 1837.

The original data are provided on a $2^\circ \times 2^\circ$ grid. In the baseline regressions, we spatially match these gridded temperature data to historical county-level administrative units. Specifically, the data constructs grid cells around each observation point to represent local

¹ For a detailed description of the temperature anomalies data, see in online introduction: <https://www.ncei.noaa.gov/pub/data/paleo/pages2k/asia2k-readme.txt>

temperature conditions (as illustrated by grids 1-6 in the figure below). We then overlay the county boundaries with the climate grids. For counties that fall entirely within a single grid cell, we directly assign the corresponding temperature anomaly. For counties that span multiple grid cells, we compute an area-weighted average based on the overlap between the county and each grid cell. This yields the annual summer temperature anomaly for each county. In a few cases, only part of a county overlaps with the climate grid. In these cases, we use the value from the overlapping grid cell. The procedure is illustrated in the figure below. Points 1-6 represent six climate observation points, and grid cells are constructed based on their locations. A, B, C, and D denote four administrative units (counties). For example, for county A, the temperature anomaly is calculated as the area-weighted average of the overlapping grid cells:

$$Temp.Anom._A = \sum_{i=1,2,4,5} \left(\frac{S_{Ai}}{S_A} \times Temp.Anom._i \right)$$

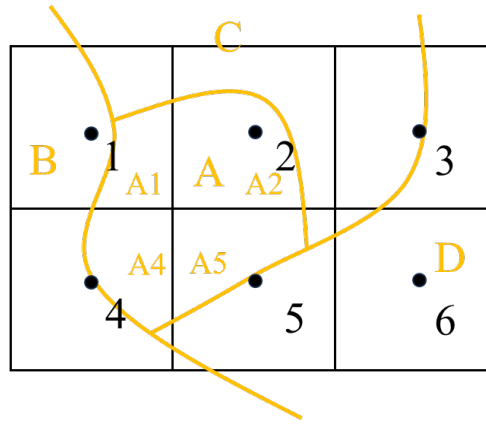


Figure A2: An example of calculation of the temperature anomalies for County A

Finally, an illustrative map of temperature fluctuations for 1837 is presented in Figure 4 of the context. In comparison, the data construction for the baseline regressions are more closely follows the structure of the original data. In addition, as a robustness check, we use alternative methods to construct climate anomalies. Although interpolation methods can be applied, they may introduce biases when grid resolution is limited. Specifically, we interpolate annual climate data using inverse distance weighting (IDW) and then match the interpolated values to administrative units based on area. The resulting measures are highly correlated with those from the baseline approach and do not change the main regression results. For example, the figure below shows the temperature anomaly for 1837 constructed using the IDW method:

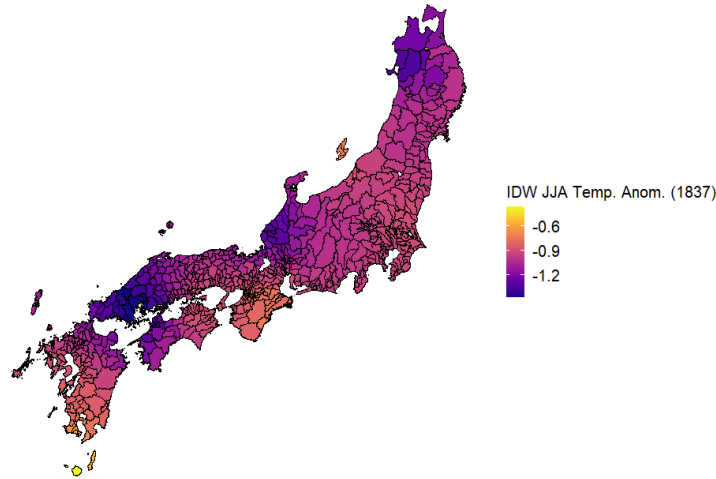


Figure A3: An example of temperature anomalies with IDW methods in 1837

For hydrological shocks, we use the Palmer Drought Severity Index (PDSI) to measure historical variations in moisture conditions. The data come from the Monsoon Asia Drought Atlas (MADA) developed by Cook et al. (2010)². Similar to the temperature reconstruction, this dataset is based on a large network of tree-ring chronologies across Asia and applies a point-by-point regression approach to reconstruct moisture conditions at individual grid points. The original data are provided on a $2.5^\circ \times 2.5^\circ$ grid. The reconstruction provides spatially explicit estimates of hydrological conditions across the Asian monsoon region. The dataset generates annual PDSI series covering the period 1300-2005. Using 1951-1989 as the calibration period, the tree-ring reconstructions are standardized and matched with instrumental observations to obtain annual moisture anomalies for each grid cell. These grid-level reconstructions serve as indicators of historical hydrological shocks. In the empirical analysis, we match this gridded PDSI data with historical county-level administrative boundaries in Japan, using the same method as described above, to calculate the annual drought and flood shock index for each county. Finally, Figure A4 provides an example of PDSI in 1837:

² For a detailed description of the PDSI data, see in online introduction: <http://drought.memphis.edu/MADA/>

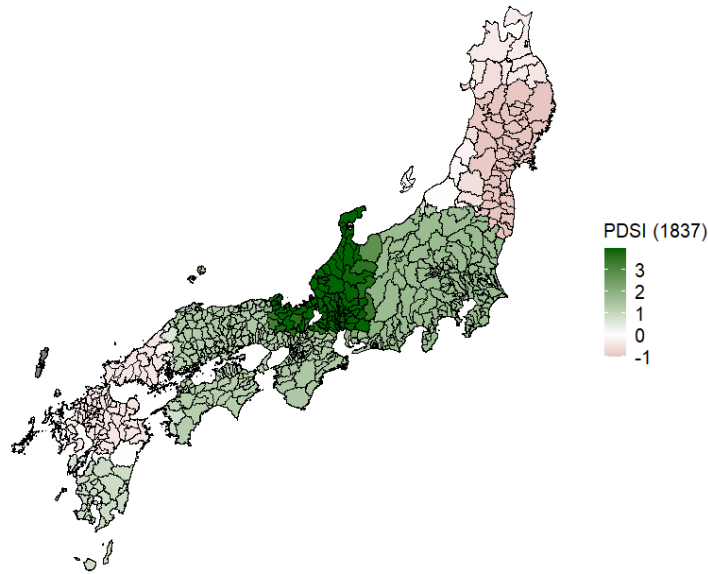


Figure A4: An example of PDSI in 1837

A.3 Control variables

We control for a range of geographic, climatic, agricultural, and socio-political factors that may influence the occurrence of uprisings. For geographic variables, we include each county's average elevation, average slope, and area (log-transformed) to capture terrain complexity and potential resource endowments. Previous studies have shown that regions with more complex terrain are more prone to conflict or rebellion, as such landscapes provide advantages for concealment and evasion (Nunn and Puga, 2012). Using data from the Global Digital Elevation Model (NASA/METI/AIST/Japan Space systems and U.S./Japan ASTER Science Team, 2001), we calculate each county's geographic characteristics.

Land productivity may also contribute to conflict (Iyigun et al., 2017; Jia, 2014). We therefore calculate the average suitability for rice cultivation in each county using data from the Food and Agriculture Organization (FAO) (IIASA/FAO, 2012). Rice being Japan's primary crop, the index ranges from 0 to 9, with higher values indicating greater suitability for rice production. The suitability index is based on a composite measure of local production conditions, including temperature, humidity, soil, and slope, measured at a 5'×5' grid resolution. Similar to the geographic variables, we take the county-level mean of rice suitability as a control variable.

We also account for socio-political factors that could confound the results. One key dimension is regional economic scale. We use the log-transformed total assessed rice output (*kokudaka*) and number of villages of each county as a proxy for economic size. Data come from records of county-level land survey and village maps in the 1830s (*Tenpō kyōchō* and

Tenpō kuniezu). The original records can be accessed at the National Archives of Japan³. Since population data during the Tokugawa period are only available at the provincial (*kuni*) level, we use 1846 provincial population figures from the (Ministry of Home Affairs, 1993) and allocate them to counties proportionally according to rice output shares.

A.5 Summary statistics

Table A1 summarizes all the variables in this paper.

Table A1 Summary statistics						
Variable	Obs.	Mean	Std.	Min	Max	Source
Popular resistances (1800-1853):						
Uprisings×100	407,844	0.117	3.891	0	400	A
Petitions×100	407,844	0.146	4.050	0	400	A
Dispersion×100	407,844	0.009	0.990	0	200	A
Disputes×100	407,844	0.199	4.665	0	400	A
Climate shocks (1800-1853):						
Temp. Anom.	407,844	-0.046	0.833	-2.656	2.968	B
PDSI	407,844	0.176	1.288	-10.524	7.917	C
Control variables (logged):						
Area	630	5.693	1.099	1.790	8.625	D
Slope	630	1.183	0.497	0	2.358	D
Elevation	630	5.159	1.068	1.208	7.210	D
Rice suitability	630	1.873	0.153	1.253	2.197	E
Rice output	630	10.526	0.812	6.949	13.194	F
Population	630	10.570	0.791	6.859	13.367	F&G
Num. of villages	630	4.398	0.895	0	7.471	H
Institutional variables:						
Normalized log distance to:						
Edo	630	0	1.000	-4.119	1.327	D
Osaka	630	0	1.000	-4.749	1.429	D
Kyoto	630	0	1.000	-3.768	1.491	D
Sendai	630	0	1.000	-5.236	1.293	D

³ For online records, see <https://www.digital.archives.go.jp/file/2549975> for land surveys and <https://www.digital.archives.go.jp/file/3146780> for village maps.

Kumamoto	630	0	1.000	-5.550	1.275	D
Territories of:						
Shogunal	630	0.227	0.419	0	1.000	I&J
Shogunal relatives	630	0.134	0.340	0	1.000	I&J
Insiders	630	0.250	0.433	0	1.000	I&J
Outsiders	630	0.382	0.486	0	1.000	I&J
Logged number of religious temples:						
Buddhism	630	1.968	1.459	0	6.370	K&L
Nichiren	630	0.269	0.623	0	4.167	K&L
Shinshū	630	0.287	0.651	0	4.934	K&L
Tendai	630	0.558	0.800	0	4.007	K&L
Shingon	630	0.989	1.067	0	5.308	K&L
Jodo	630	0.591	0.867	0	3.970	K&L
Zenshū	630	1.202	1.229	0	5.598	K&L
Nantoro	630	0.025	0.207	0	3.045	K&L
Shinto shrines	630	1.619	1.525	0	6.399	K&L
Other variables:						
Log distance to main road	630	4.003	4.545	0	11.771	M
Harbor	630	0.126	0.405	0	4.000	M
Log distance to coast	630	5.295	4.921	0	11.547	D
Mine	630	0.046	0.238	0	2.000	N
Coal mine	630	0.051	0.472	0	7.000	N
Towns in 1872	630	0.630	0.521	0	3.178	O
Rice price of 13 markets	702	1.139	0.397	0.510	3.190	P

Data source: A: Aoki (1971); B: Cook et al. (2013); C: Cook et al. (2010); D: Digitalized map; E: Fao (2012); F: *Tenpō kyōchō*; G: Ministry of Home Affairs (1993); H: *Tenpō kuniezu*; I: Kimura (1995); J: Kodama and Kitajima (1989); K: Ando (1977); L: Ando(1980); M: Doboku Gakkai (1936); N: Ohashi (1909); O: Sozeiryō (1873); P: Iwahashi (1981).

Appendix B. Robustness checks

B.1 Alternative measurements

In the baseline specification, climate shocks are measured by summer temperature anomalies at the county-year level, capturing deviations from the long-term mean within a fixed-effects panel estimation. In Table B1, we instead focus on short-term climate fluctuations. Specifically, we construct an alternative measure defined as:

$$\Delta Temp. Anom._{it} = Temp. Anom._{it} - Temp. Anom._{it-1}$$

$$\Delta PDSI_{it} = PDSI_{it} - PDSI_{it-1}$$

Here, the variable captures deviations relative to the previous year, thereby reflecting short-run changes in climate conditions. The regression results show that temperature shocks remain significantly negative, indicating that a decline in temperature relative to the previous year leads to a higher probability of uprisings. In contrast, the coefficients on PDSI are generally insignificant. Although the estimate in column 3 is statistically significant, its magnitude is small. A one-unit increase in PDSI relative to the previous year is associated with only a 5.9 % decrease in the probability of uprisings, compared to a mean of 0.103. Given that most PDSI values lie within the range of -4 to 4, this effect is economically negligible.

Furthermore, in the baseline estimation in the main text, we use a binary indicator to capture whether any uprising occurs in a given county-month, in order to account for potential within-period correlation in events. As an additional robustness check, Table B2 and B3 use the count of uprisings within each month as the dependent variable. The results are consistent with the baseline results. Also, in Table B4, we aggregate the number of uprisings at the annual level. The results remain consistent with the baseline findings: a decline in summer temperature within a given year is associated with a significant increase in the total number of uprisings in that year.

We further address the potential influence of the Tenpō Crisis. One concern is that our results may be driven by the famine and social unrest of the 1830s, a period characterized by repeated cold shocks and excessive rainfall that led to severe agricultural shortfalls and frequent uprisings. In Table B5, we assess this concern in two ways. First, we estimate the regressions separately for the periods before and after 1830. Second, we exclude the Tenpō Crisis years (1832-1839) from the sample. The results remain highly consistent with the baseline estimations, and we find no evidence of systematic differences across subsamples. These findings indicate that our results are not driven by this extreme historical episode, but instead reflect a more general and stable relationship between climate shocks and social instability. This strengthens the external validity of our results and suggests that the estimated effects are not confined to periods of exceptional crisis.

B.2 Dynamic regressions

We further conduct a series of dynamic regressions. First, we examine the dynamic effects of climate shocks by including both lagged and lead terms of temperature anomalies in the regression. The results, reported in Figure B1, show that uprisings are primarily driven by contemporaneous temperature shocks. The coefficients on both lagged and lead terms are statistically insignificant, suggesting no strong evidence of anticipatory or persistent effects.

In addition, since our baseline climate measure captures summer temperature anomalies, Figure B2 explores seasonal heterogeneity by estimating the effects on uprisings across different months within the same year. The results indicate that summer temperature shocks have no significant impact on uprisings occurring in winter (January, February, and December)

or spring (March to May). In contrast, the effects are significantly negative for uprisings in summer and fall, consistent with the agricultural calendar. This suggests that temperature shocks matter most during the core growing and harvesting seasons.

Finally, in Figure B3, we regress uprisings in early months (January-February and March-May) on lagged summer temperature anomalies from the previous year. The estimates are again insignificant. Taken together, these results indicate that the impact of temperature shocks on social unrest is concentrated within the same agricultural cycle, particularly during the growing and harvest seasons. This finding is consistent with the historical context discussed in the main text, where lower temperatures reduce agricultural output and increase the likelihood of uprisings.

B.3 Alternative constructions of temperature anomalies

The paleotemperature data from Cook et al. (2013) are originally provided in a gridded format. However, given the limited spatial resolution of historical climate reconstructions, temperature values may exhibit discontinuities across adjacent grid cells. To address this concern, as described in Appendix A.2, we apply an inverse distance weighting (IDW) interpolation method to construct smoothed estimates of temperature anomalies at the county-year level, which we use as an alternative measure of climate shocks. The regression results, reported in Table B6, are broadly consistent with the baseline findings. This suggests that concerns related to the spatial resolution of the paleoclimate data do not materially affect our main results.

Table B1 Effects on changes in climate conditions

	Uprisings $\times$ 100		
	(1)	(2)	(3)
Δ Temp. Anom.	-0.0414*** (0.011)		-0.0425*** (0.011)
Δ PDSI		-0.00539 (0.004)	-0.00616* (0.004)
County FE	YES	YES	YES
Year-month FE	YES	YES	YES
Controls $\times$ Year FE	YES	YES	YES
Observations	407592	407592	407592
Adjusted R^2	0.010	0.010	0.010

Note: Dependent variable in column 1-4 is the number of uprisings in each county each month from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table A1. All columns include county fixed effects and year-month fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

<0.1.

Table B2 Effects on changes in climate conditions

	Num. of Uprisings $\times$ 100		
	(1)	(2)	(3)
Temp. Anom.	-0.0583*** (0.015)		-0.0576*** (0.015)
PDSI		0.0117	0.0111
County FE	YES	YES	YES
Year-month FE	YES	YES	YES
Controls $\times$ Year FE	YES	YES	YES
Observations	407592	407592	407592
Adjusted R^2	0.012	0.012	0.012

Note: Dependent variable in column 1-4 is the number of uprisings in each county each month from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table A1. All columns include county fixed effects and year-month fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table B3 Effects on changes in climate conditions

	Num. of Uprisings $\times$ 100		
	(1)	(2)	(3)
Δ Temp. Anom.	-0.0404*** (0.013)		-0.0418*** (0.013)
Δ PDSI		-0.00711 (0.004)	-0.00787* (0.004)
County FE	YES	YES	YES
Year-month FE	YES	YES	YES
Controls $\times$ Year FE	YES	YES	YES
Observations	407592	407592	407592
Adjusted R^2	0.012	0.012	0.012

Note: Dependent variable in column 1-4 is the number of uprisings in each county each month from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table A1. All columns include county fixed effects and year-month fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

<0.1.

Table B4 Annual data

	Num. of uprisings $\times$ 100		
	(1)	(2)	(3)
Temp. Anom.	-0.699*** (0.177)		-0.691*** (0.177)
PDSI		0.140 (0.094)	0.133 (0.094)
County FE	YES	YES	YES
Year-month FE	YES	YES	YES
Controls $\times$ Year FE	YES	YES	YES
Observations	33966	33966	33966
Adjusted R^2	0.056	0.056	0.056

Note: Dependent variable in column 1-4 is the number of uprisings in each county each year from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table A1. All columns include county fixed effects and year fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table B5 Exclusion of Tenpo crisis

	Num. of uprisings $\times$ 100			
	Before 1830s	After 1830s	Exclude Tenpo Crisis	During Tenpo Crisis
	(1)	(2)	(3)	(4)
Temp. Anom.	-0.0668*** (0.018)	-0.0475* (0.028)	-0.0362** (0.015)	-0.182** (0.073)
PDSI	0.0170** (0.008)	-0.00605 (0.013)	0.0132** (0.006)	0.0447 (0.066)
County FE	YES	YES	YES	YES
Year-month FE	YES	YES	YES	YES
Controls $\times$ Year FE	YES	YES	YES	YES
Observations	226440	181152	354756	52836
Adjusted R^2	0.011	0.015	0.009	0.023

Note: Dependent variable in column 1-4 is the number of uprisings in each county each year from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table A1. All columns include county fixed effects and year fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table B6 Temperature anomalies with IDW method

	Uprisings $\times 100$		Num. of uprisings $\times 100$	
	(1)	(2)	(3)	(4)
Temp. Anom. (IDW)	-0.0880*** (0.025)	-0.0877*** (0.025)	-0.0970*** (0.030)	-0.0966*** (0.030)
PDSI		0.0108* (0.006)		0.0115 (0.008)
County FE	YES	YES	YES	YES
Year-month FE	YES	YES	YES	YES
Controls $\times$ Year FE	YES	YES	YES	YES
Observations	407592	407592	407592	407592
Adjusted R^2	0.014	0.014	0.012	0.012

Note: Dependent variable in column 1-4 is the number of uprisings in each county each year from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table A1. All columns include county fixed effects and year fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

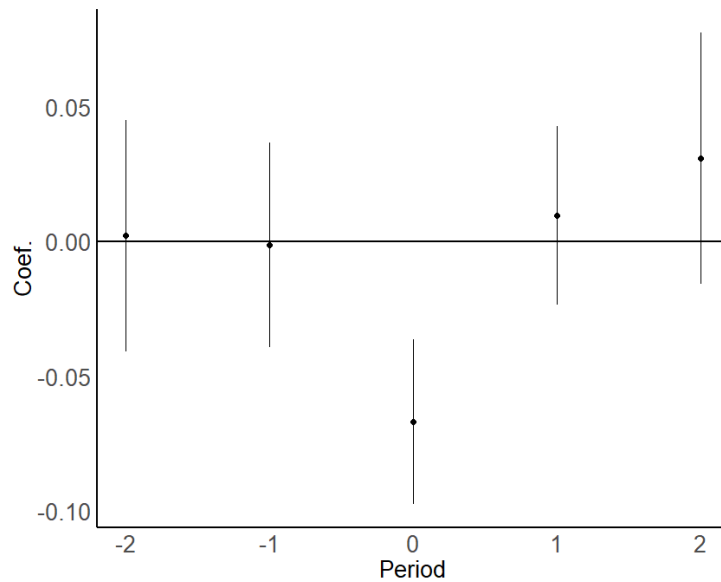


Figure B1 Dynamic effects of temperature anomalies

Note: Dependent variable in the figure is the number of uprisings in each county each month from 1800-1853. The independent variables are lagged and future terms of temperature anomalies in each year in each county. The set of control variables is reported in Table A1. All columns include county fixed effects and year-month fixed effects. The standard errors are

clustered at the county level. The points indicate estimations of the coefficients, with the gray lines indicate the 95% confidence intervals.

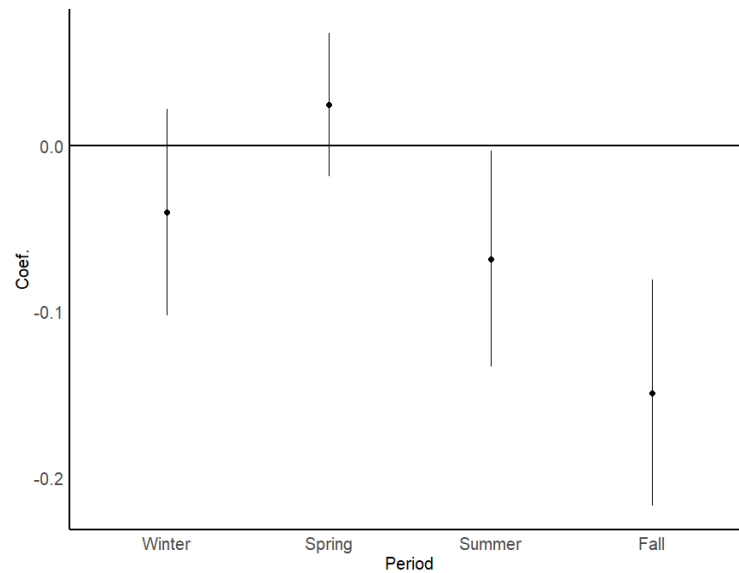


Figure B2 Dynamic effects of seasonal heterogeneity

Note: Dependent variable in the figure is the number of uprisings in each county each month from 1800-1853, separated by different seasons (Winter: Jan., Feb., Dec.; Spring: Mar. - May; Summer: Jun. - Aug.; Fall: Sept. - Nov.). The independent variable is temperature anomalies in each year in each county. The set of control variables is reported in Table A1. All columns include county fixed effects and year-month fixed effects. The standard errors are clustered at the county level. The points indicate estimations of the coefficients, with the gray lines indicate the 95% confidence intervals.

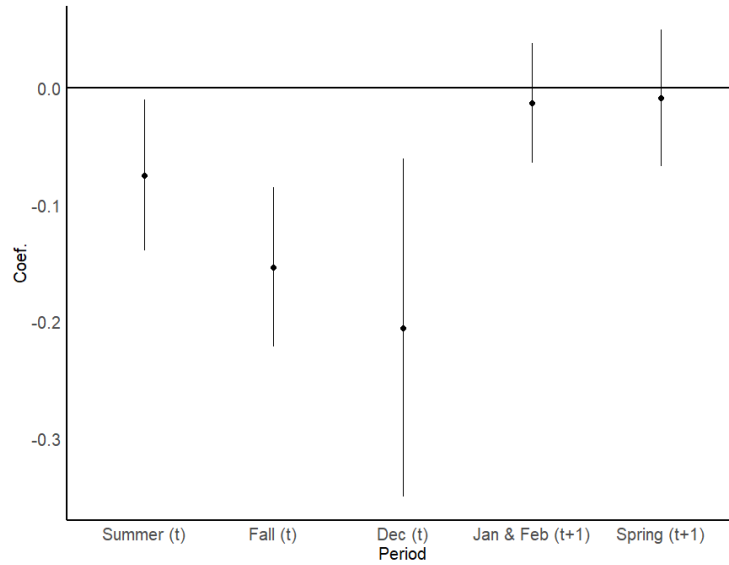


Figure B3 Dynamic effects of seasonal heterogeneity with lagged terms

Note: Dependent variable in the figure is the number of uprisings in each county each month from 1800-1853, separated by different seasons (Spring: Mar. - May; Summer: Jun. - Aug.; Fall: Sept. - Nov.). The independent variables for Jan. & Feb., and Spring is lagged temperature anomalies in each year in each county, while others are temperature anomalies in each year. The set of control variables is reported in Table A1. All columns include county fixed effects and year-month fixed effects. The standard errors are clustered at the county level. The points indicate estimations of the coefficients, with the gray lines indicate the 95% confidence intervals.

Appendix C. Further discussions

This section provides additional evidence related to the discussion in Section 5.3 of the main text. In Table C1, we examine whether alternative economic channels mentioned in the existing literature, such as transportation accessibility and non-agricultural income opportunities, played a mitigating role in pre-modern Japan.

In column 1, we use the logged distance from each county to major roads as a moderating variable (data sources: Doboku Gakkai, 1936; and the same applies below). The underlying hypothesis is that areas with better transport access could facilitate interregional grain trade, thereby alleviating the adverse effects of local climate shocks. In column 2, we use the number of mines (Ohashi, 1909) within a county as a proxy for alternative employment opportunities, capturing whether non-agricultural income sources could offset agricultural losses. In columns 3 and 4, we examine the logged distance to the coast and the presence of a port (Doboku Gakkai, 1936) to measure the potential role of maritime transport. Finally, in column 5, we use the logged number of towns (*machi*) within each county (Sozeiryo, 1873) to proxy for the degree of urbanization, reflecting the availability of handicraft and commercial activities that could provide alternative income sources under climate stress.

The regression results, reported in Table C1, show that the interaction terms between these variables and temperature shocks are not statistically significant. This suggests that alternative economic channels did not play a meaningful mitigating role in pre-modern Japan, despite some degree of decentralization and occupational diversification. Instead, the results support the interpretation that the impact of climate shocks operated primarily through agricultural output losses and the resulting increases in food prices, which in turn led to famine and social instability.

Finally, in Table C2, following the estimations in Table 4, we match each county to its nearest major grain market to examine the effect of grain price fluctuations on popular resistance. The results for different types of resistance are reported across columns.

Column 1 shows that increases in grain prices significantly raise the incidence of uprisings. In particular, higher grain prices in the nearest market during the same year are associated with a marked increase in collective unrest. At the same time, the coefficient on temperature anomalies remains significantly negative and is broadly comparable in magnitude to the results reported in column 6 of Table 2. Combined with the findings in Table 9, which show that cold shocks increase grain prices, these results suggest that rising grain prices constitute an important channel through which climate shocks translate into social unrests, but not the only one. For rural societies, cold shocks could reduce agricultural output, trigger food shortages, and increase subsistence pressure, all of which may directly contribute to unrest beyond the price mechanism.

In addition, column 2 shows that increases in grain prices are also associated with a rise in petitioning activities. In contrast, the relationship between climate shocks and petitions is not statistically significant. This pattern suggests that petitions were more likely to emerge as a response to market-based economic stress, whereas uprisings were more closely linked to broader subsistence crises caused by climatic shocks.

Table C1 Substitute economical channels

Uprisings $\times 100$					
	(1)	(2)	(3)	(4)	(5)
Temp. Anom.	-0.0566*** (0.014)	-0.0553*** (0.013)	-0.0555*** (0.013)	-0.0571*** (0.016)	-0.0533*** (0.016)
Temp. Anom. $\times$ Dist. to main road	-0.000179 (0.002)				
Temp. Anom. $\times$ Mine		-0.0323 (0.038)			
Temp. Anom. $\times$ Harbor			-0.0143 (0.020)		

Temp. Anom. ×				-0.0000204	
Dist. to coast				(0.002)	
Temp. Anom. ×					-0.00592
Num. of Town					(0.016)
County FE	YES	YES	YES	YES	YES
Year-month FE	YES	YES	YES	YES	YES
Controls × Year FE	YES	YES	YES	YES	YES
Observations	407592	407592	407592	407592	407592
Adjusted R^2	0.014	0.014	0.014	0.014	0.014

Note: Dependent variable in column 1-5 is the uprisings in each county each month from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table B1. All columns include county fixed effects and year-month fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table C2 Rice price and popular resistances

Resistances × 100	Uprisings	Petitions	Dispersions	Disputes
	(1)	(2)	(3)	(4)
Rice price	0.177** (0.083)	0.238*** (0.080)	-0.0114 (0.010)	0.0725 (0.077)
Temp. Anom.	-0.0579*** (0.013)	-0.0227 (0.018)	-0.000983 (0.005)	0.0214 (0.017)
PDSI	0.0103 (0.006)	-0.0108 (0.007)	0.000636 (0.002)	-0.0115 (0.008)
County FE	YES	YES	YES	YES
Year-month FE	YES	YES	YES	YES
Prov. trend	YES	YES	YES	YES
Controls × Year FE	YES	YES	YES	YES
Observations	407592	407592	407592	407592
Adjusted R^2	0.014	0.010	0.003	0.013

Note: Dependent variable in column 1-5 is the uprisings in each county each month from 1800-1853, as indicated at the top of each column. The set of control variables is reported in Table B1. All columns include county fixed effects and year-month fixed effects. The standard errors shown in parentheses are clustered at the county level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Appendix D. Additional discussions of Lord identity

In Section 5 of the main text, we classify different types of domain authority. Building on the discussion in Appendix A.1, we rely on the data compiled by Kimura (1995), which records the

identity of its ruling authority as well as the scale of rice outcomes in each village, measured in *koku* (a standard unit of rice yield), for each village within a county prior to the Meiji Restoration in the 1860s. Although the Tenpo land survey records provide tax data that are closer in time to our study period (as the control variable in the estimations, see in A.3), they do not contain systematic information on village-level lordship. Moreover, between 1800 and the Boshin War in 1866-1868, the identity of domain lords remained largely stable, with limited large-scale reassignments.

Based on this information, we are able to identify the dominant ruling authority for each county by aggregating village-level affiliations. Furthermore, by combining these data with historical records on individual domains, we can classify each lord according to their political status and their relationship with the shogunate.

For illustration, Figure D1 presents a simplified example. Suppose a county contains 25 villages, and each village has an equal level of agricultural rice output (normalized to 1). These villages are governed by different feudal lords, denoted as A, B, and C. In some cases, a single village may be jointly associated with multiple lords, as shown in row 4, column 5 of the example. Based on this structure, we can aggregate village-level outcomes to identify the dominant lord within the county, defined as the one controlling the largest share of total output (in this example, lord A).

At the same time, each lord can be classified into different political categories, including shogunate lands, insider lords, outsider lords, and shogunal relatives. Using historical records on lord classifications Kodama and Kitajima (1989), we are able to assign a political identity to the dominant lord in each county. The resulting classification of dominant lord types at the county level is reported in Figure D2.

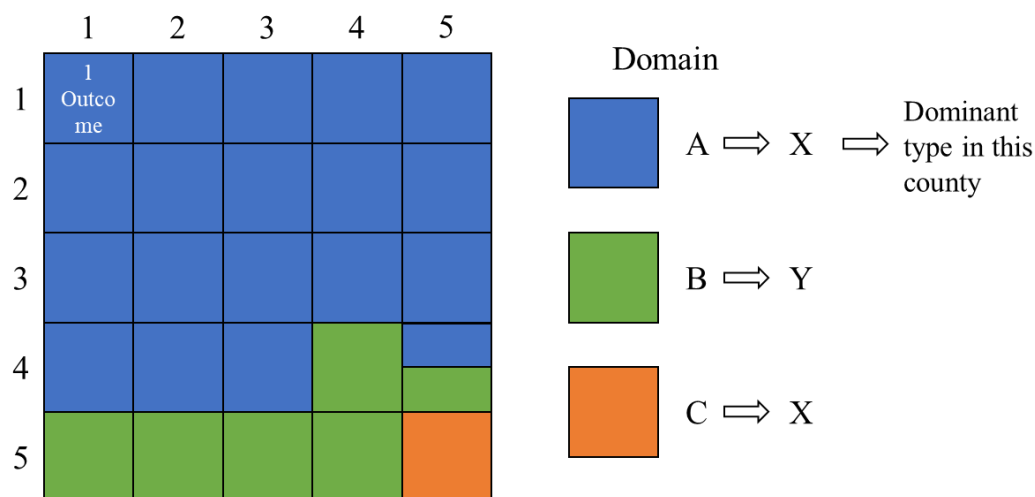


Figure D1: An example of lord identity in each county

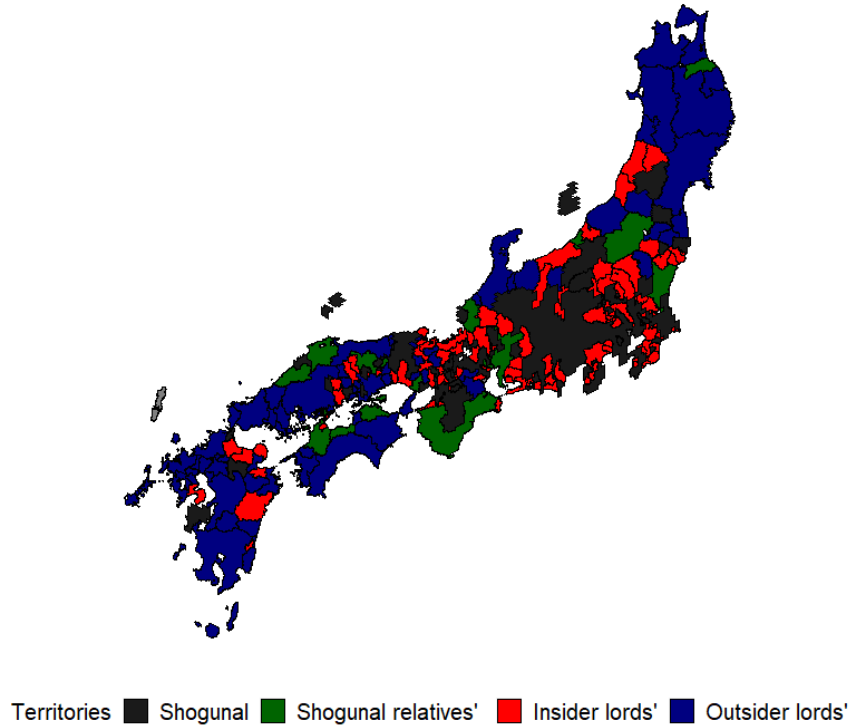


Figure D2: Spatial distribution of territories

Appendix E. Additional historical backgrounds of Japanese Buddhism

In this section, we provide a brief overview of the historical development of major Buddhist sects in Japan and their relationships, in order to complement the findings presented in Tables 7 and 8. Overall, Japanese Buddhism is predominantly rooted in the Mahayana (*Daijō*) tradition, and its sectarian development can be broadly classified into four major categories: Nara Buddhism, Esoteric Buddhism, Pure Land Buddhism, and Zen Buddhism. A series of scholarship has examined the historical evolution of Japanese Buddhism (e.g., Ruppert and Deal, 2015; Matsuo, 2007; Osumi and Nakao, 1998; Watanabe, 1958). Figure E1 plots the distribution of the dominant Buddhist sect, defined as the sect with the largest number of temples within each county. The data are described in Section 5 of the main text. The figure shows that different sects are distributed across regions rather than concentrated in specific areas, providing relatively exogenous variation for identifying the effects of different sects.

In addition, we briefly outline the historical background and relationships among different sects, which helps us assess which sects are more likely to function as informal institutions in mitigating the effects of climate shocks. Figure E2 illustrates the timing of the introduction or establishment of different Buddhist sects in Japan, as well as their relationships.

1. Nara Buddhism (results are reported in column 6 of Table 8 in the main text and

hereinafter). Nara Buddhism refers to the early institutionalized forms of Buddhism established in Japan during the Nara period (710-794), largely under strong state patronage. It includes major scholastic schools which were closely associated with the imperial court and aristocratic elites. These sects emphasized doctrinal study, monastic discipline, and the preservation of canonical texts, rather than active engagement with lay communities. As a result, their organizational structures were concentrated in large temple complexes and lacked deep penetration into rural society. While Nara Buddhism played a foundational role in shaping the intellectual and institutional framework of Japanese Buddhism, its capacity for grassroots mobilization and local social coordination remained limited.

2. Tendai sect (Column 1 in Table 7). The Tendai sect originated in the early Heian period and was founded by Saichō (767-822). Tendai incorporated a wide range of doctrinal traditions, including elements of esoteric Buddhism introduced from China, and thus developed a comprehensive religious system. As shown in the Figure D1, several Buddhist sects that emerged during the Kamakura period (1185-1333) maintained historical links with the Tendai tradition. Therefore, in Table 7, we first examine the influence of Tendai. However, unlike Shingon, esoteric ritual in Tendai remained only one component within a broader doctrinal framework. Tendai institutions were primarily centered on large temple complexes such as Enryakuji on Mount Hiei, and their activities often focused on doctrinal training and elite religious practice. Although Tendai temples sometimes exercised political and military influence in the medieval period, their organizational structure was relatively concentrated and less embedded in village-level society. As a result, their capacity for sustained grassroots mobilization in response to local shocks was comparatively limited.

3. Shingon sect (Column 2 in Table 7). The Shingon sect, founded by Kukai (774-835), emphasized esoteric rituals aimed at protecting the state, ensuring agricultural prosperity, and warding off disasters. Shingon temples frequently conducted rainmaking ceremonies, epidemic-control rituals, and prayers for harvest stability. These practices were often embedded in local governance arrangements, with temples cooperating with domain authorities and village leaders during periods of environmental stress.

4. Jōdo sect (Column 3 in Table 7). The Jōdo sect was founded by Honen (1133-1212) and emphasized salvation through faith in Amida Buddha. Its teachings spread widely among commoners and contributed to the formation of local religious communities. However, compared with Shinshū, Jōdo sect organization tended to remain relatively loose and centered on devotional practice rather than structured collective organization. While Jōdo temples interacted with local society and maintained village-level presence, they did not typically develop dense lay networks or institutionalized mechanisms for collective mobilization. As a result, although Jōdo sects had a broad social base, their capacity to coordinate large-scale responses to economic shocks remained limited.

5. Shinshū sect (Column 4 in Table 7). The Shinshū sect, founded by Shinran (1173-1263), developed from the Jōdo tradition but placed greater emphasis on lay participation and secular

life. Followers were not required to withdraw from family or occupational activities, which facilitated the formation of temple-centered local communities. These communities often developed dense horizontal networks across villages, supporting collective action and mutual assistance.

6. Nichiren sect (Column 5 in Table 7). The Nichiren sect, founded by Nichiren (1222-1282), emphasized the relationship between religious order and social stability. Nichiren interpreted disasters and social unrest as consequences of moral and political disorder, encouraging engagement with public affairs. The sect spread through preaching and local associations, forming organized lay networks in both urban and rural areas.

7. Zenshū sect (Column 1-4 in Table 8). The Zenshū tradition, including schools such as Soto Zen and Rinzai Zen, emphasized meditation and introspective practice as the primary path to enlightenment. Zen institutions often focused on monastic discipline and temple-based training rather than outreach to the general population. Although some Zen temples-maintained ties with samurai elites and regional authorities, their organizational structures were typically centered on monastic communities rather than village-level networks. As a result, their role in coordinating collective responses to environmental shocks was relatively limited compared with more socially embedded sects.

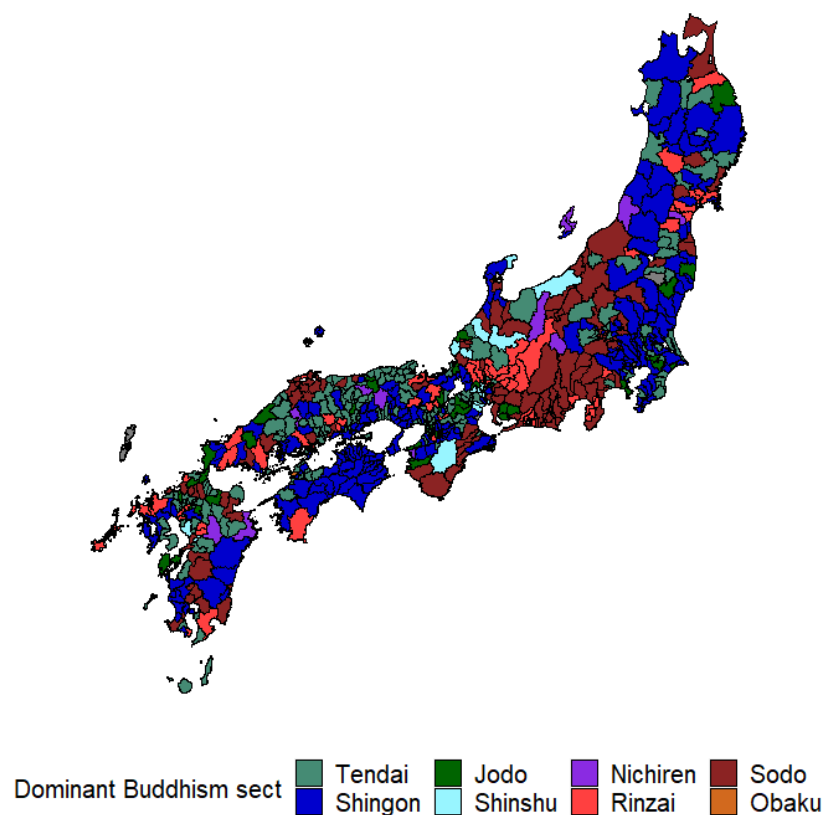


Figure E1: Distribution of dominant Buddhist sects during late Tokugawa period

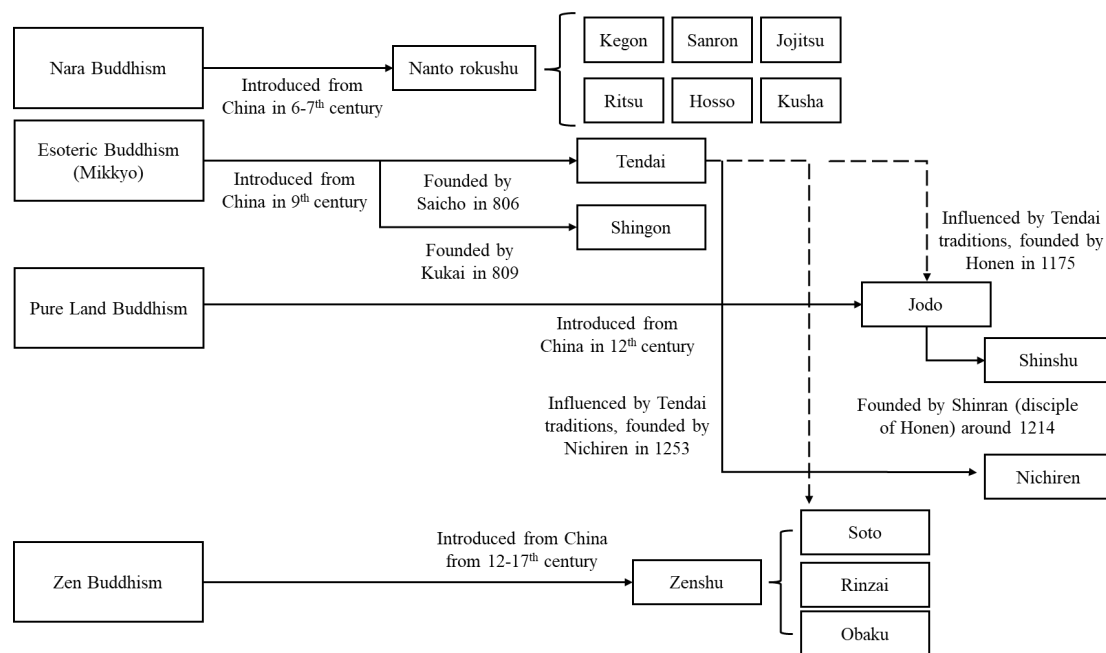


Figure E2 History and relationships of Japanese Buddhist sects

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